

Task 1: String, Array and Arithmetic operations

- Concept: Strings and String Manipulation (.trim(), .substring(), .toUpperCase())
- Goal: Capture a messy text string from the keyboard, clean it, and format it properly.

```
package studenttracker;

import java.util.Scanner;

public class Task1_stringsArrayUserInput {
    public static void main(String[] args) {
        Scanner inputScanner = new Scanner (System.in);

        // === Task 1: String Input & Cleaning ===
        System.out.println("=== Task 1: String Input & Cleaning ===");
        System.out.print("Enter student name (e.g., ' john doe '): ");
        String rawName = inputScanner.nextLine();

        String cleanName = rawName.trim();
        String formattedName = cleanName.substring(0, 1).toUpperCase() +
cleanName.substring(1);
        System.out.println("Saved Name: '" + formattedName + "'");

        // === Task 2: Parsing & Casting ===
        System.out.println("\n=== Task 2: Parsing & Casting ===");
        System.out.print("Enter text grade for Math: ");
        String mathText = inputScanner.next();
        System.out.print("Enter text grade for Science: ");
        String scienceText = inputScanner.next();

        int mathScore = Integer.parseInt(mathText);
        int scienceScore = Integer.parseInt(scienceText);
        double initialAverage = (double) (mathScore + scienceScore) / 2;
        System.out.println("Initial 2-Class Average: " + initialAverage + "%");
    }
}
```

```
// === Task 3: Arrays & Loops ===
System.out.println("\n=== Task 3: Arrays & Loops ===");
System.out.print("How many total classes do you want to track? ");
int totalClasses = inputScanner.nextInt();

int[] scheduleGrades = new int[totalClasses];
int arraySum = 0;

for (int i = 0; i < scheduleGrades.length; i++) {
    System.out.print("Enter text grade for class #" + (i + 1) + ": ");
    String textGrade = inputScanner.next();

    // Reuse Task 2 wrapper parsing directly inside our array tracking loop
    scheduleGrades[i] = Integer.parseInt(textGrade);
    arraySum += scheduleGrades[i];
}

double finalAverage = (double) arraySum / scheduleGrades.length;
System.out.println("Final Schedule Average: " + finalAverage + "%");

inputScanner.close();
}
}
```